

STEM BIO-AI Evidence-Surface Scan v1.7.9

Repository: jensseck/bio | Commit: 100a0b67497e | Branch: main | Audit Date: 2026-05-31 | Mode: LOCAL_ANALYSIS | Policy: default (mirror_only)

48 / 100

Final Score

T1 Quarantine

Use Scope:
Exploratory review only; no patient-adjacent use.

Weighted model: Stage 1 x 0.40 + Stage 2R x 0.20 + Stage 3 x 0.40 - Risk Penalty = 48

Stage 1 README Evidence	Stage 2R Repo-Local Consistency	Stage 3 Code / Bio Responsibility	Stage 4 Replication Evidence
75 / 100	40 / 100	25 / 100	30 / 100

Code Integrity PASS C1 Credentials No direct credential patterns detected by local CLI scan. WARN C2 Dependency Pinning External operational dependency signal surfaced in code-integrity lane. PASS C3 Deprecated Paths No deprecated patient-adjacent metadata patterns detected. PASS C4 Exception Handling No executable fail-open exception handler detected. WARN C5 Compliance Boundary Unsupported legal/compliance claim surfaced in boundary-integrity lane. WARN C6 Mock Auth Boundary Mock-auth or auto-login boundary surfaced in code-integrity lane.	Remediation Targets - Clinical-adjacent surfaces exist without an explicit non-diagnostic/non-clinical boundary. - Self-asserted compliance or privacy-governance claim requires independent verification. - Legal, privacy, or compliance claim appears without supporting governance or security-grounding evidence in re - Core workflow appears materially dependent on named external service providers; local or self-host claims may Positive Evidence - Package metadata was available for repo-local consistency checks. AIRI Risk Triggers - Covered Risks: 7 / 32 Rate: 0.219 24.01.03 Safe... why: C5_compliance_b...
Bio Deterministic Diagnostics SMILES Surface Integrity not_detected=1 SMILES RDKit Validation not_detected=1 SMILES Parser Guard not_detected=1 Silent Mock Fallback not_detected=1 Traceability Manifest Surface not_detected=1 Bio Subprocess Run Trace not_detected=1 Regulatory basis note Aligned to current official source classes as of May 2026: EU AI Act (Regulation (EU) 2024/1689), FDA QMSR, FDA AI-enabled device guidance themes, and IMDRF SaMD/GMLP frameworks. This is a traceability aid, not a compliance or clearance determination. Traceability summary: Structural signals partially align with traceability scaffolding. This remains a pre-audit traceability aid, not a compliance determination.	

Stage 1 — README Evidence Signal | Weight: 0.40

75 / 100

S1 Score

Check	Points	Evidence / Finding
Baseline	+60	Non-nascent README evidence baseline.
BIO/medical terms in README	+10	README exposes bio/medical domain vocabulary.
BIO/medical terms in package	+5	Package metadata exposes bio/medical domain vocabulary.
R2: Regulatory Framework	+5	Self-asserted privacy/compliance language detected without stronger regulatory-framework evidence. [partial-credit ladder: +15 strong framework +5 weak self-asserted...
R3: Clinical Boundary	-5	CA-INDIRECT surface lacks explicit non-clinical or non-diagnostic boundary.

Calculation: 60 plus Stage 1 evidence additions/deductions = 75

• Clinical-Adjacent: **YES** (CA-INDIRECT) • Explicit Disclaimer: **ABSENT** • T0 Hard Floor: **Clear**

Stage 2R — Repo-Local Consistency | Weight: 0.20

40 / 100

S2R Score

Check	Points	Evidence / Finding
Baseline	+60	Non-nascent local repository baseline. — Every repository that is not nascent starts at 60. This baseline accounts for basic structural maturity. —...
R2R-1: README / Package Alignment	+15	README has domain overlap with package metadata or entry points. — README and package metadata share bio-domain vocabulary, indicating claim-to-implementation alignment. —...
R2R-D2: Missing Clinical Boundary (PENALTY)	-20	Clinical-adjacent surfaces exist without an explicit non-diagnostic/non-clinical boundary. — Clinical-adjacent repository lacks an explicit 'research use only' or 'not for...
R2R-D4: Unsupported Workflow Claim (PENALTY)	-15	README/docs claim runnable workflow, CLI, test, or demo support without matching local support surfaces. — README or docs describe runnable workflow support that local...

60 plus local consistency additions/deductions = 40 **Local Contradiction / Insufficient Consistency**

Stage 3 — Code & Bio Responsibility | Weight: 0.40

25 / 100

S3 Score

Engineering Accountability (T-series)

Check	Points	Evidence / Finding
T1: CI/CD Workflow	0 / 15	No workflow files detected. — CI/CD workflows (GitHub Actions, GitLab CI, CircleCI) verify that commits do not silently break the pipeline. Full credit (15) requires...
T2: Domain-Specific Tests	0 / 15	No tests detected. — Domain-specific tests verify biological outputs — e.g., sequencing pipeline correctness, variant call validation, or genomic data integrity. Full credit...
T3: Changelog & Release Hygiene	0 / 15	No changelog detected. — A CHANGELOG tracks which version fixed which defect — essential for regulatory traceability and reproducibility audits. CHANGELOG.md, CHANGELOG, or...

Biological Integrity (B-series)

Check	Points	Evidence / Finding
B1: Data Provenance Controls	15 / 15	Dependency manifest detected with data source, IRB, or dataset citation language. — Dependency manifests (requirements.txt, pyproject.toml, environment.yml) establish...
B2: Bias / Limitations Documentation	0 / 15 [Manual review required]	No bias/limitations language detected by local CLI scan. — Documentation of algorithmic bias, limitations, or model boundary conditions. Score 8 for boundary language; max...
B3: COI & Funding Disclosure	5 / 5	COI, funding, sponsor, or acknowledgement language detected. — Conflict of interest and funding disclosure in README or FUNDING.md. Required for institutional review context...

Stage 3 Gap Analysis — Path to Next Tier

- **T-series (engineering) attained:** 0 / 45 **B-series (bio integrity) attained:** 20 / 35
- **Local CLI scan maximum:** 55 / 100 (T1+T2+T3 max 15 each; B1 max 10; B2/B3 require manual review)
- **Gap to T3 (final score >= 70):** 22 points needed across all stages
- **Gap to T4 (final score >= 85):** 37 points needed across all stages
- **B2 Bias/Limitations:** Not detectable — requires manual audit of README, model card, or supplementary documentation for validation boundaries and algorithmic limitations
- **B3 COI/Funding:** Not detectable — requires inspection of README or FUNDING.md for conflict of interest and funding source disclosure

Stage 4 — Replication Evidence Lane | Separate lane

30 / 100

S4 Score

Check	Points	Evidence / Finding
S4: Container / Runtime Environment	10 / 10	Container or compose file exists.
S4: Reproduce Target	0 / 10	No Makefile detected.
S4: Environment Lock Evidence	10 / 10	Environment, dependency, or lock manifest detected.
S4: Exact Dependency Pins / Hashes	10 / 10	Exact dependency pin or hash evidence detected.
S4: Reproducibility Section	0 / 10	README exists but no reproducibility or replication section heading was detected.
S4: Checksums / Integrity Files	0 / 10	No evidence detected for S4_checksum_files.
S4: Dataset / Data Source URL	0 / 10	Documentation exists but no dataset URL or data source URL was detected.
S4: Model Artifact URL / Checksum	0 / 10	Documentation exists but no model artifact URL/checksum evidence was detected.
S4: CITATION.cff	0 / 5	No evidence detected for S4_citation_cff.
S4: License / Use Restriction	0 / 0	No license/use restriction language detected.
S4: CLI Entrypoint	0 / 5	No package metadata or Python AST surface detected.
S4: Deterministic Seed Setting	0 / 5	No deterministic seed setting detected.
S4: Runnable Examples	0 / 5	No evidence detected for S4_runnable_examples.
stage_4_raw_total	30 / 100	Raw Stage 4 rubric total. Stage 4 is reported separately and does not alter final score.

Replication tier: R1. Stage 4 is reported separately and does not alter the formal score.

Final score remains **48 / 100** (T1 Quarantine) even when Stage 4 moves. This lane exists to show reproducibility and operational evidence posture separately from the formal repository score.

Code Integrity — Deep Analysis

Check	Points	Evidence / Finding
C1: Hardcoded Credentials	PASS	No direct credential patterns detected by local CLI scan. Scan: Scans for AWS access keys (AKIA*), OpenAI keys (sk-*), GitHub tokens (ghp_*), and api_key = '...' patterns...
C2: Dependency Pinning	WARN	External operational dependency signal surfaced in code-integrity lane. Scan: Checks whether requirements.txt / pyproject.toml / environment.yml use exact version pins...
C3: Deprecated Patient Paths	PASS	No deprecated patient-adjacent metadata patterns detected. Scan: Scans deprecated/directories for patient metadata patterns: patient_id, patient_age, patient_sex...
C4: Fail-Open Exceptions	PASS	No executable fail-open exception handler detected. Scan: Detects fail-open exception patterns: 'except Exception: pass' or 'except: return True' in code — these silently...
C5: Compliance Boundary Integrity	WARN	Unsupported legal/compliance claim surfaced in boundary-integrity lane. Scan: Detects unsupported legal/compliance claims or clinical-boundary weaknesses in reviewed...
C6: Mock Auth / Fail-Open Boundary	WARN	Mock-auth or auto-login boundary surfaced in code-integrity lane. Scan: Detects mock-auth, auto-login, or no-auth local/self-host boundary patterns in README, config, and...

Remediation Guidance

- [WARN] C2: Dependency Pinning:**
→ Pin all dependencies to exact versions (== for pip, hash-pinning for conda). Run pip-audit or safety regularly. Consider pip-compile for reproducible lock files. Unpinned ranges in clinical-adjacent pipelines create silent regression risk.
- [WARN] C5: Compliance Boundary Integrity:**
→ Treat privacy, legal, or clinical-adjacent claims as governance obligations. If README or product text invokes HIPAA, compliance, or self-hosted clinical safety, surface supporting controls, operating boundaries, and deployment constraints explicitly.
- [WARN] C6: Mock Auth / Fail-Open Boundary:**
→ Do not present self-host, local-mode, or privacy-sensitive flows as production-like if they rely on mock authentication, auto-login, or no-auth convenience boundaries. Separate demo convenience from trust posture.

Classification & Repository Analysis

Check	Points	Evidence / Finding
Clinical Adjacent	YES	Severity: CA-INDIRECT. Triggered by BIO/CLINICAL_OUTPUT term regex match across README, docs, and code.
T0 Hard Floor	Clear	No T0_HARD_FLOOR condition detected.
Explicit Disclaimer	ABSENT	Disclaimer pattern not found in README or docs. High impact on Stage 1 and Stage 2R scores.
Files Scanned	166	Total files indexed by recursive walk. Text files only for content analysis; binary files counted but not read.
Execution Mode	LOCAL_ANALYSIS	No LLM calls. No network access. No runtime execution. Deterministic regex + file-system scan only.

File Integrity (SHA-256)

File	SHA-256 Hash
README.md	199862D708D85AF0B126FD4129E5F134D6E9E804F6F8249F940F3DA16DC190AA

Priority Improvement Roadmap

Priority 1: Clinical-adjacent surfaces exist without an explicit non-diagnostic/non-clinical boundary.

→ Add a prominent 'Research Use Only — Not for Clinical or Diagnostic Use' disclaimer to README H1 or H2 section. Reference applicable frameworks: FDA SaMD guidance, EU AI Act Article 6, or IRB oversight requirements for your deployment context.

Priority 2: Self-asserted compliance or privacy-governance claim requires independent verification.

→ Review this finding and implement appropriate controls before supervised or clinical-adjacent deployment.

Priority 3: Legal, privacy, or compliance claim appears without supporting governance or security-grounding evidence in reviewed repository sources.

→ Review this finding and implement appropriate controls before supervised or clinical-adjacent deployment.

Priority 4: Core workflow appears materially dependent on named external service providers; local or self-host claims may overstate operational independence.

→ Review this finding and implement appropriate controls before supervised or clinical-adjacent deployment.

Priority 5: External operational dependency signal surfaced in code-integrity lane.

→ Review this finding and implement appropriate controls before supervised or clinical-adjacent deployment.

Priority 6: Unsupported legal/compliance claim surfaced in boundary-integrity lane.

→ Review this finding and implement appropriate controls before supervised or clinical-adjacent deployment.

Priority 7: Mock-auth or auto-login boundary surfaced in code-integrity lane.

→ Review this finding and implement appropriate controls before supervised or clinical-adjacent deployment.

Positive Evidence Summary

- Package metadata was available for repo-local consistency checks.

AIRI Risk Triggers Summary

Covered Risks: **7 / 32** | Coverage Rate: **0.219** | Bundle Scope: **curated_medical_clinical_subset**

Markdown, HTML, PDF, and explain outputs may compact repeated same-file evidence rows or narrative blocks. Refer to JSON for full per-finding detail.

- **24.01.03** Safe exploration problem with widely deployed AI assistants — primary: C5_compliance_boundary_integrity — why:

C5_compliance_boundary_integrity: Unsupported legal/compliance claim surfaced in boundary-integrity lane.

- **24.04.01** Physical and Psychological Harms — primary: C2_dependency_pinning — why: C2_dependency_pinning: External operational dependency signal surfaced in code-integrity lane.

- **33.01.05** Privacy and security — primary: C2_dependency_pinning — why: C2_dependency_pinning: External operational dependency signal surfaced in code-integrity lane.

Known gaps preview: 65.03.03 Reidentification, 70.02.02 Misinformation — hallucination of clinical knowledge

Method Boundary

Deterministic local CLI scan. No LLM, network, or runtime test execution is required.

Scope boundary: Runtime behavior, model output correctness, dynamic validation, wet-lab reproducibility, and clinical validation are outside the scope of this local CLI scan. This report assesses structural signals only.

Report Metadata	
Field	Value
Schema Version	stem-ai-local-cli-result-v1.6
STEM BIO-AI Version	1.7.9
Generated (local date)	2026-05-31
Report Validity	180 days from audit date
Execution Mode	LOCAL_ANALYSIS
Repository	yorkeccak/bio
Remote URL	https://github.com/yorkeccak/bio.git
Branch	main
Commit (HEAD)	100a0bf7497e62ead024df34d8c2e00ae74b8d99
Files Scanned	166
Final Score / Tier	48 / 100 — T1 Quarantine

Independent audit summary — STEM BIO-AI v1.7.9 | Not clinical certification. Not regulatory clearance. Not medical advice.